

SAMIULLAH

Senior AI Engineer · Production ML Systems · Multi-Agent & Data Infrastructure

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PROFESSIONAL SUMMARY

AI Engineer with 2+ years designing, deploying, and scaling production AI/ML systems for real-world business use cases. Sole backend architect for a production content-automation platform (Revnix/rext.ai) and lead architect on ConnectXeo, a multi-agent Shopify platform coordinating 7 subagents across 40+ live tool integrations. Strong foundation in SQL/NoSQL databases (PostgreSQL, Redis, MongoDB), data pipeline design, and workflow orchestration for reliability under production load. Published open-source work (langctl on PyPI, 14+ releases) defining reusable agent orchestration and middleware standards. Comfortable owning architecture end-to-end, building pipeline evaluation and monitoring processes, and reviewing design decisions across teams.

TECHNICAL SKILLS

AI / Agentic Systems: LangGraph, LangChain, Multi-Agent Orchestration, Coordinator-Subagent Patterns, Tool-Aware Routing, Middleware Architecture, Human-in-the-Loop Design, Agent Architecture Review

Machine Learning & Deep Learning: PyTorch, Transformers (Hugging Face), Parameter-Efficient Fine-Tuning (PEFT/LoRA), Computer Vision (YOLO, ResNet, VGG16), Traditional ML, Hyperparameter Tuning/Optimization

Data & Databases: PostgreSQL, Redis, MongoDB, pgvector, SQL, Alembic (schema migrations), Semantic Search, RAG (Multi-Query Expansion, Query Decomposition, RAG Fusion, Reranking)

Pipelines & Orchestration: Async workflow orchestration (Redis-based), multi-stage pipeline design (discovery → scoring → generation → approval → publishing), production reliability engineering

Backend, Infrastructure & MLOps: Python (4 yrs, lead), FastAPI, Docker, AWS (EC2, ECR), MLflow (Model Tracking), DVC (Data Version Control), GitHub Actions (CI/CD), Multi-Tenant Deployment

Frontend: Next.js, React, TypeScript

PROFESSIONAL EXPERIENCE

AI/ML Engineer — [Revnix \(rext.ai\)](https://revnix.com) March 2025 – Sep 2026

Stack: LangGraph · OpenAI GPT · FastAPI · PostgreSQL · Redis · Docker

- Sole backend architect for a production content-automation platform, designing scalable multi-stage AI/ML pipelines: automated topic discovery → relevance scoring → LLM-based outline generation → human-in-the-loop approval → SEO-optimized publishing.
- Engineered async state persistence with PostgreSQL and Alembic-managed schema migrations, plus Redis-based workflow orchestration, to keep pipelines reliable under production traffic.
- Owned end-to-end deployment and reliability of AI/ML services running continuously in production, working directly with SQL data stores to support downstream reporting and analytics.

Lead AI Agent Architect — [ConnectXeo \(connectxeo.com\)](https://connectxeo.com) 2025 – Present

Stack: FastAPI · LangGraph · Next.js · PostgreSQL · pgvector · WhatsApp Cloud API · Docker

- Architected a multi-agent system from scratch — coordinator-subagent pattern with 7 specialized agents, each with isolated tool scopes, error boundaries, and retry logic across 40+ live Shopify API integrations.
- Defined and enforced architectural standards for agent-to-agent communication, state management, and tool-call routing, producing reusable patterns adopted across every subsystem on the platform.
- Built multi-tenant infrastructure with per-merchant data isolation, shared orchestration patterns, and multi-channel deployment (WhatsApp Business Cloud API, web chat, REST API) — solving real production scalability challenges.
- Reviewed agent and system design decisions across multiple domains (orders, refunds, inventory, draft orders), ensuring each subsystem scales independently behind consistent contracts.

AI Engineer — [Emulxion](https://emulxion.com) Jan 2024 – Nov 2024

Stack: PyTorch · FastAPI · React · Docker

- Designed a production RAG architecture using multi-query expansion, query decomposition, fusion, and intelligent reranking — established the retrieval pattern later adopted across the engineering team.
- Built and deployed a real-time inference pipeline serving live production traffic via FastAPI and React with sub-second response times.

OPEN SOURCE & KEY PROJECTS

langctl — Production Agent CLI (pypi.org/project/langctl)

- Built and published an open-source CLI (pip install langctl) that scaffolds, runs, and deploys production LangChain/LangGraph agents — 14+ releases, CI/CD via GitHub Actions, Trusted Publishing on PyPI, Apache-2.0 licensed.
- Designed a semantic middleware pipeline (guardrails → context → limits → reliability → human-in-the-loop → capability → custom) with extensible custom classes.
- Engineered built-in long-term memory (SQLite/Postgres), semantic search, 25-provider model support, and zero-config deployment via Docker Compose with automatic HTTPS (Caddy).

AgentGenesis — Agent-to-Agent Framework

- Building an open-source framework where agents configure and spawn other agents — plugin-based architecture for composable, reusable agent patterns at scale.

ODS-M — Middleware-Controlled Autonomous Search Agents

- Published research on autonomous web search agents with tool-aware middleware routing — outperformed baseline benchmarks on multi-hop reasoning tasks.

EDUCATION

BS Computer Science — University of Haripur | 2022 – 2026 | CGPA: 3.5 / 4.0

ACHIEVEMENTS & PUBLICATIONS

- 1st Place — ZABIST AI Cup 2024
- 1st Place — Thrive Pakistan
- Published: ODS-M — Middleware-Controlled Autonomous Web Search Agents